PROJECT HANDOVER DOCUMENTATION

Squato-Meter

A Bar-Mounted Wearable for Squat Form Analysis

Using Sensor Fusion and Machine Learning

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Course

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https://github.com/seb2233/SquatoMeter | https://github.com/seb2233/SquatoMeterApp

1 Project Overview

The **Squato-Meter** is a compact wearable device that clamps onto the outer sleeve of a standard Olympic barbell. It combines an IMU (accelerometer + gyroscope) with a Time-of-Flight depth sensor to capture real-time biomechanical data during barbell squats — measuring bar tilt, squat depth, bar path, and repetition count — without restricting the lifter's movement.

Sensor data is filtered on-board with a Kalman filter, transmitted wirelessly via **Bluetooth Low Energy (BLE)** to a laptop or an app.

Said mobile app receives and displays data in real time.

2 Source Code

Both repositories and a **user manual** are on GitHub under `seb2233`:

> **Firmware & ML pipeline**
> `https://github.com/seb2233/SquatoMeter`
> **Companion mobile app**
> `https://github.com/seb2233/SquatoMeterApp`

3 Hardware Components

Component	Part
MCU	Seeed Studio XIAO ESP32S3
IMU	SparkFun BMI270 (Acc. + Gyro.)
Depth sensor	Pimoroni VL53L5CX (ToF)
PCB	Custom-designed board
Enclosure	3D-printed housing + velcro mount
Power	Rechargeable LiPo battery (in case)

The device mounts on the **outer bar sleeve**, which is mechanically decoupled from the inner shaft via bearings — meaning the Squato-Meter stays rotationally fixed while the bar rotates under the lifter's grip.

4 Data Streams

Signal	Rate	Unit
Accel. X / Y / Z	100 Hz	m/s^2
Gyro. X / Y / Z	100 Hz	deg/s
Roll (Kalman)	100 Hz	deg
ToF depth	30 Hz	mm

5 Known Issues

> **Critical — Low Battery Voltage Bug**
>
> When the battery voltage drops too low, the power supply to the **accelerometer and gyroscope (BMI270 IMU) becomes insufficient**, causing both sensors to stop functioning correctly. The device may appear to be running but will produce no valid IMU data.
>
> **Fix:** Physically remove the battery from the enclosure and reinsert it to force a full power reset. Recharge the battery before the next session.
>
> **Recommendation for future work:** Add a low-voltage warning (LED or BLE alert) and/or a low-dropout voltage regulator to ensure stable power delivery to sensors at lower charge levels.

6 Task Tracker (Jira)

All sprint work, backlog items, and open issues are tracked on the project Jira board:

> **Jira Board (SM project)**
> `https://racething.atlassian.net/jira/software/projects/SM/boards/2/backlog`
> Issue SM-54 and surrounding tickets contain the most recent work items. Request access from the project supervisor if needed.

7 Recommended Next Steps

1. **Fix the battery/voltage issue** — add a hardware-level power indicator or regulator
2. **Add further features** like a machine learning algorithm into the phone application

8 Key Contacts

Role	Person
Author	Sebastian Gal
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